

INTRINSIC BIAS DIFFERENTIAL BETWEEN VACANCY LOOPS AND INTERSTITIAL LOOPS

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Point-defect migration into an infinitesimal edge dislocation loop in an isotropic linear elastic medium has been studied. Particular care has been taken to include the effects of the saddle-point shape anisotropy of the point defect. The reaction radii and biases are derived and found to depend on the loop nature and the point-defect shape at the saddle point. Application of the theory to copper and iron shows that the bias differential between vacancy and interstitial loops can be very large.

1. Introduction

One of the most important features of the microstructure of crystalline materials subjected to particle irradiation is the presence of dislocation loops. To understand the mechanisms governing the macroscopic effects of irradiation damage, such as void swelling, irradiation creep and growth, knowledge concerning the evolution of the dislocation loops is essential. In this regard, the study of point-defect diffusion into dislocation loops has received a great deal of attention [1–7].

It is generally perceived that dislocation loops are biased sinks that attract interstitials more strongly than vacancies. Under most circumstances, in the absence of sinks of higher bias, the bias of dislocation loops causes a net flow of interstitials to them. In fact, this is generally accepted to be the basic mechanism responsible for void swelling [8,9], and has also been proposed as a mechanism for irradiation creep and growth [10,11]. However, the net flow of interstitials to the dislocation loops will result in the growth of interstitial loops and the shrinkage of vacancy loops. Although this conclusion has been supported by observation in the case of stainless steel, it is contrary to observations with titanium [12], and zirconium [13] and its alloys [14]. Here, the presence of large vacancy loops, whose radii increased with dose, showed that vacancy-loop growth is possible in some materials. Bullough, Hayns and Woo [1] made a first attempt to resolve this apparent contradiction between theory and experiment. These authors proposed that the bias increases with increasing loop size, smaller

loops being relatively neutral sinks. Thus, provided that voids are absent and grain boundaries are far away, there will be a net flow of vacancies to the smaller loops. This competing process provides a sufficient condition for the simultaneous growth of both kinds of loops. Subsequent work [4–6], in which the drift-diffusion problem is treated more rigorously, came to essentially the same findings. A difficulty with this proposed explanation, however, is that there is no experimental evidence that the vacancy loops are smaller than the adjacent interstitial loops, as expected from the theory. This discrepancy prompts an alternate approach that searches for a mechanism that would give an intrinsic bias differential between vacancy loops and interstitial loops. However, it has been shown [6] that if the drift field is harmonic there should be no appreciable difference between the sink strength of a vacancy loop and that of an interstitial loop. Within the confines of linear elasticity theory and a point defect modelled as an isotropic center of dilatation or contraction, the point-defect/dislocation-loop interaction seems to be restricted to a harmonic form.

An alternative theory was suggested by Wolfer and Si-Ahmed [7]. They proposed that non-linear elasticity theory may produce an intrinsic bias differential between vacancy and interstitial loops. Because of the inherent difficulty of calculations involving non-linear elasticity theory, these authors, in their calculation of the capture efficiencies of the loops, used an “apparent Burgers vector” obtained by fitting the total loop volume change (associated with the loop) to the non-linear

elastic results. Since non-linear elasticity effects usually produce an additional volume expansion, the apparent Burgers vector is larger than the true one for interstitial loops and smaller for vacancy loops. This causes a difference in the capture efficiency of the two loop types even if they have the same true Burgers vector. A flaw in this approach is that, despite the correct loop volume, there is no guarantee that the correct strain field of the loop will be even approximately reproduced. This is an important point because it is the detailed nature of the dislocation strain field which ultimately determines the form and magnitude of the bias. In the present case, an increase (decrease) in the apparent Burgers vector causes more (less) dilatation in the dilatational region, and more (less) compression in the compressive regions of the strain field of the interstitial (vacancy) loop. This is inconsistent with computer simulation results [15,16] which indicate that, at least for straight edge dislocations, non-linear effects introduce additional dilatations in *both* the dilatational *and* compressive regions. If these results are extrapolated to dislocation loops, they mean an increase from the linear elastic results in the dilatation and a decrease in the compression for *both* types of loops. From this point of view, then, it is not clear how non-linear elasticity theory can produce a significant bias differential as these authors have claimed.

Another alternative which we are adopting here is to look at the anisotropy of the point-defect strain field. In the above calculations, the effect of the point-defect/sink interaction on the point-defect current into the sink is described by the continuum theory of drift-diffusion. Within this approximation, the point-defect current can be written as

$$\mathbf{j} = -D_0 \nabla \rho - D_0 \beta \rho \nabla E \quad (1)$$

where D_0 is the ideal (i.e., in the absence of the drift field) diffusion coefficient, β is the reciprocal of the product of the Boltzmann constant and the absolute temperature, ρ is the point-defect density and E the drift field (the interaction field between the point defect and the sink). In metals, the main contribution to E comes from the elastic interaction between the strain field of the sink and that of the point defect, which is usually modelled as an isotropic expansion (or contraction) center.

Recently, Dederichs and Schroeder [17] have pointed out that the continuum description given by eq. (1) is, in general, inadequate. By adopting a microscopic approach they showed that, firstly, the effect of the drift field on the point-defect mobility was not accounted for, and secondly, there was no clear indication of whether E referred to the point defect in its equilibrium

or its saddle-point configuration. They did find that if the saddle-point configuration of the point defect (or the dipole tensor for elastic interaction) is isotropic, then the form of eq. (1) can be retained by replacing D_0 with $D = D_0 \exp[-\beta(E_s - E_g)]$ and E by E_g , where E_s is the drift field when the point defect is in the saddle-point configuration and E_g the corresponding quantity at the equilibrium position. Note that both E_s and E_g are functions of the point-defect position $\mathbf{r}$. Then it can be shown that the flux is determined solely by E_s , and all the previous results can then be kept, simply by replacing E with E_s and ρ with $\rho \exp[-\beta(E_s - E_g)]$ [18,19].

However, saddle-point configurations of point defects, particularly those of vacancies, are not isotropic in general. This is intuitively evident and has been shown to be true in cubic metals by computer simulation results [20,21]. Depending on the extent of the anisotropy of the interacting fields, results obtained by modelling the point defect simply as a center of isotropic expansion (or contraction) could be very much in error, and important effects might be concealed.

In the present case, the anisotropy of the saddle-point configurations of point defects required the inclusion of the shear component in the point-defect strain field, which interacts with the shear component of the dislocation-loop strain field. It will be shown that this additional interaction is different for the two kinds of loops. As a result, a difference in bias exists.

The purpose of this paper is to calculate the point-defect sink strengths and the bias of an infinitesimal edge dislocation loop (IEDL), taking into account, explicitly, the anisotropy of the saddle-point configuration. We consider only the case of no external stress. Analytic expressions for the reaction radii as a function of temperature, sink densities, loop nature, and point-defect dipole tensor will be derived. It will be shown that there is an intrinsic difference between the biases of a vacancy-type IEDL and an interstitial-type IEDL. The difference is a function of the shapes of the vacancies and interstitials at the saddle-point configuration. The theory is then applied to fcc copper and bcc iron where computer simulation results for the elastic dipole-force tensors for the point defects are available. It will be shown that this bias differential is significant in both cases.

2. Formulation

The general problem for point-defect drift-diffusion, under the action of a slowly varying (spatially) applied strain ξ_{kl} , and including the effect of the saddle-point

anisotropy, has been formulated from a microscopic approach by Dederichs and Schroeder [17]. According to these authors, drift-diffusion can be described in terms of the renormalized concentration $W = \rho \exp(\beta E_g)$ by the extremum principle of the Lagrangian

$$L[W] = \int \frac{d^3\mathbf{r}}{V} [\tilde{D}_{ij}(\mathbf{r}) \partial_i W(\mathbf{r}) \partial_j W(\mathbf{r}) - p(\mathbf{r}) W(\mathbf{r})], \quad (2)$$

where V is the integration volume, ∂_i denotes $\partial/\partial r_i$, $p(\mathbf{r})$ is the production rate and $\tilde{D}_{ij}$ is the renormalized diffusion tensor, given by

$$\tilde{D}_{ij} = \frac{1}{2} \sum_{\hat{h}} h_i h_j \lambda_{\text{eff}}^{0(\hat{h})} \exp[\beta \xi_{kl} P_{kl}^s(\hat{h})]. \quad (3)$$

Here, the summation is over all nearest-neighbour jump directions $\hat{h}^*$, h_i is the i th component of the nearest-neighbour position vector to which a jump occurs, $P_{kl}^s(\hat{h})$ is the elastic dipole-tensor (i.e., the double force tensor) of the point defect at the saddle point in the jump direction $\hat{h}$, and $\lambda_{\text{eff}}^{0(\hat{h})}$ is an effective (averaged over different non-equivalent equilibrium configurations of the point defect) ideal lattice-jump frequency in the $\hat{h}$ direction. Here, as in the rest of the paper, we sum over repeated indices. Furthermore, we note that W in eq. (2) refers to the total concentration of non-equivalent equilibrium configurations [17] of the same defect. For a cubic defect in a cubic crystal, we can neglect the $\mathbf{h}$ dependences of $\lambda_{\text{eff}}^{0(\hat{h})}$ and the nearest-neighbour jump distance. Then we may write eq. (3) as

$$\hat{D}_{ij} = D_0 \frac{3}{Z} \sum_{\hat{h}} \hat{h}_i \hat{h}_j \exp[\beta \xi_{kl} P_{kl}^s(\hat{h})], \quad (4)$$

where D_0 is the ideal diffusion coefficient and Z the co-ordination number. In non-cubic crystals, or non-cubic defects, the same assumption leads to an approximation where the anisotropy of the diffusion tensor, resulting from the anisotropy of the crystal lattice, is averaged out. In this approximation, D_0 is replaced by the averaged ideal diffusion coefficient D_0^{eff} , which is effectively what is used in eq. (1) in most of the previous calculations [1–7]. This approximation is not expected to cause large errors in normal cases where the anisotropy of lattice diffusion is not severe. Furthermore, since the present study concentrates on drift-diffusion in the presence of an applied stress, which mainly concerns the exponential term in eq. (4), one may hope that the above approximation does not change, at least, the qualitative picture.

* The symbol $\hat{\cdot}$ in $\hat{h}$ denotes the unit vector in the direction of $\mathbf{h}$. This symbol is used throughout with similar meaning.

A further approximation necessary to make the solution of this problem tractable is the spherical symmetrization procedure first introduced by Schroeder and Dettman [19] and used successfully by Woo [5] for dislocation loops. Within this approximation, we used a spherical symmetric ansatz for the renormalized density $W(\mathbf{r}) \simeq n_0(r)$. The extremum principle of the Lagrangian then gives the following one-dimensional equation:

$$\frac{1}{r^2} \frac{d}{dr} \left[r^2 \tilde{D}_{\text{eff}}(r) \frac{dn_0}{dr} \right] + p(r) = 0, \quad (5)$$

where $\tilde{D}_{\text{eff}}(r)$ is given by

$$\tilde{D}_{\text{eff}}(r) = \int \frac{d\Omega}{4\pi} \hat{r}_i \tilde{D}_{ij} \hat{r}_j. \quad (6)$$

Here $d\Omega$ is an elemental solid angle and $\hat{r}_i$ is the i th component of the unit vector $\hat{\mathbf{r}}$.

To simplify the notation, we denote $\xi_{kl} P_{kl}^s(\hat{h})$ in eq. (4) by $-E_s(\mathbf{r}; \hat{h})$. Then eqs. (4) and (6) imply

$$\tilde{D}_{\text{eff}}(r) = 3D_0^{\text{eff}} Z^{-1} \sum_{\hat{h}} \int \frac{d\Omega}{4\pi} (\hat{\mathbf{r}} \cdot \hat{\mathbf{h}})^2 \exp[-\beta E_s(\mathbf{r}; \hat{h})] \quad (7)$$

$$\simeq 3D_0^{\text{eff}} \int \frac{d\Omega_{\hat{h}}}{4\pi} \int \frac{d\Omega}{4\pi} (\hat{\mathbf{r}} \cdot \hat{\mathbf{h}})^2 \exp[-\beta E_s(\mathbf{r}; \hat{h})] \quad (8)$$

if we assume that the jump directions are isotropic and approximate the summation by an integration. Then if we define

$$\langle f(\mathbf{r}) \rangle_{\hat{\mathbf{r}}} \equiv \int \frac{d\Omega_{\hat{\mathbf{r}}}}{4\pi} f(\mathbf{r}) \quad (9)$$

eq. (8) can be written as

$$\tilde{D}_{\text{eff}}(r) = 3D_0^{\text{eff}} \langle (\hat{\mathbf{r}} \cdot \hat{\mathbf{h}})^2 \exp(-\beta E_s) \rangle_{\hat{\mathbf{r}}}. \quad (10)$$

In the following, we denote $\langle (\hat{\mathbf{r}} \cdot \hat{\mathbf{h}})^2 \exp(-\beta E_s) \rangle_{\hat{\mathbf{r}}}$ by $\langle \exp(-\beta E_s) \rangle$. We further define $\bar{E}_s(r)$ by

$$\exp(-\beta \bar{E}_s) = 3 \langle \exp(-\beta E_s) \rangle; \quad (11)$$

then eq. (10) becomes

$$\tilde{D}_{\text{eff}}(r) = D_0^{\text{eff}} \exp(-\beta \bar{E}_s). \quad (12)$$

If we denote the reaction radius by R_a , the capture radius of the sink by r_a , and the sink-volume (i.e., the total volume divided by the number of sinks) radius by R , we have the following results.

(a) *Constant boundary concentration method.* In this method, the sink is surrounded by a spherical reservoir with radius R , centred at the sink, and maintained at a constant concentration $\bar{C}$. The current into the sink is calculated, and the reaction radius is related to the

current. From this method, R_a is given by

$$R_a = \left\{ \int_{r_0}^R t^{-2} \exp[\beta \bar{E}_s(t)] dt \right\}^{-1} \quad (13)$$

from eq. (28) of Woo [5] by putting $\bar{K}_L = 0$.

(b) *Effective medium approximation.* In this case, the production of point defects is considered, in contrast to (a). Sufficient documentation is found in the literature for this method [22] that it is not necessary to describe it here. Using eq. (39) of Woo [5], the reaction radius is given by

$$R_a = \left\{ \int_{r_0}^R (t^{-2} - t/R^3) \exp[\beta \bar{E}_s(t)] dt \right\}^{-1}. \quad (14)$$

Note that, firstly, the reaction radii calculated from the effective medium are always larger than those from the constant boundary concentration and, secondly, the difference between the two decreases as R increases (or the sink density decreases), so that there is no difference in the limit when the sinks are very dilute.

With the reaction radius calculated, the rate theory can then be invoked to obtain the microstructure evolution and macroscopic effects. Thus, the problem is reduced to the evaluation of $\bar{E}_s$. As pointed out by Schroeder and Dettmann [19], the integrands in eqs. (13) and (14) are important only for large values of t , because $\bar{E}_s(t)$ has a large negative value for small values of t , as can be seen from the definition of $\bar{E}_s(r)$ in eq. (11). Under such circumstances, as shown in Woo [23], we can write, to a good approximation,

$$\beta \bar{E}_s(r) \approx 3\langle \beta E_s \rangle - \frac{3}{2}\langle \beta^2 E_s^2 \rangle + \frac{9}{2}\langle \beta E_s \rangle^2. \quad (15)$$

To evaluate $\bar{E}_s$, we use the displacement field of an IEDL given by Kroupa [24], and expand the point-defect dipole tensor P_{kl}^s in terms of its eigenvector $\hat{e}^{(\sigma)}$ ($\sigma = 1, 2, 3$), i.e.,

$$P_{kl}^s = P_{\sigma} \hat{e}_k^{(\sigma)} \hat{e}_l^{(\sigma)} \quad (16)$$

where $P = 1/3 \text{Tr} P_{kl}^s$ and p_{σ} is a normalized eigenvalue such that P_{σ} is the σ th eigenvalue. We note that this implies

$$\sum_{\sigma} P_{\sigma} = 3. \quad (17)$$

The interaction energy $E_s = -P_{kl}^s e_{kl}$ can then be evaluated [23]:

$$E_s = -\frac{k_0 P Q}{r^3} \left\{ (2\nu - 1)(3 - 2p_{\sigma} \hat{b}_{\sigma}^2) + 3[(1 - 2\nu)p_{\sigma} \hat{r}_{\sigma}^2 + 4\nu \hat{b}_{\alpha} \hat{r}_{\alpha} p_{\sigma} \hat{b}_{\sigma} \hat{r}_{\sigma} + 3\hat{b}_{\alpha} \hat{b}_{\beta} \hat{r}_{\alpha} \hat{r}_{\beta}] - 15p_{\sigma} \hat{r}_{\sigma}^2 \hat{b}_{\alpha} \hat{b}_{\beta} \hat{r}_{\alpha} \hat{r}_{\beta} \right\}, \quad (18)$$

where $k_0 = -[8\pi(1 - \nu)]^{-1}$, ν being the Poisson ratio, and

$$Q = \pm \delta A b \begin{cases} + \text{sign for vacancy loops} \\ - \text{sign for interstitial loops,} \end{cases}$$

δA being the loop area and b the magnitude of the Burgers vector. $\hat{r}_j$ and $\hat{b}_j$ are the j th component of the unit position vector of the point defect and the unit Burgers vector of the IEDL, respectively. In eq. (18) we have used the coordinate system in which the $\hat{e}^{(\sigma)}$ are basis vectors. If we put

$$P = \kappa \Delta V = \frac{2}{3} \frac{1 + \nu}{1 - 2\nu} \mu \frac{4\pi}{3} r_0^3 (3\epsilon) \quad (19)$$

and

$$Q = \pi r_L^2 b \quad (20)$$

where κ is the bulk compressibility, ΔV is the relaxation volume, μ is the shear modulus, r_0 the free defect radius in continuum theory, ϵ the radial strain as the defect is placed in the elastic medium, and r_L the loop radius, eq. (18) reduces in the isotropic limit, i.e., with $p_{\sigma} = 1$ for all σ , to

$$E_s(p_{\sigma} = 1) = \frac{2}{3} \frac{1 + \nu}{1 - 2\nu} \mu r_0^3 \pi \epsilon r_L^2 b (1 - 3 \cos^2 \theta) / r^3. \quad (21)$$

This is the exact formula usually used for the point-defect/dislocation-loop interaction (see e.g., Bullough et al. [25]) in continuum theory calculations.

In a recent paper, Bullough et al. [6] showed that if E_s is harmonic, i.e., $\partial_i \partial_i E_s = 0$, then there is no appreciable difference between the point-defect current into a vacancy or an interstitial loop of equal size. In this connection, it can be shown that the point-defect/loop interaction in eq. (18) is not, in general, harmonic.

In fact, from eq. (18), we obtained [23]

$$\partial_i \partial_i E_s = -\frac{k_0 P Q}{r^5} (p_{\sigma} - 1) \left\{ 12 \hat{b}_{\sigma}^2 - 30 \hat{r}_{\sigma}^2 + 15 \left[14 \hat{r}_{\sigma}^2 (\hat{r}_{\alpha} \hat{b}_{\alpha})^2 - 8 \hat{b}_{\alpha} \hat{b}_{\beta} \hat{r}_{\sigma} \hat{r}_{\beta} \right] \right\}. \quad (22)$$

Eq. (22) clearly shows that, unless $p_{\sigma} = 1$ for all σ , $\partial_i \partial_i E_s$ is non-zero. This result immediately points to the possibility of a difference between the point-defect current into loops of different character (i.e., vacancy or interstitial), thus giving rise to a bias difference between the two. This difference is intrinsic and is entirely a consequence of the anisotropy of the point-defect dipole tensor at the saddle point. In the following, we evaluate the energy $\bar{E}_s$ and the corresponding reaction radius R_a . From this, we will then be able to examine more closely the details, the magnitude, and the consequences of this difference.

3. Evaluation of $\beta\bar{E}_s$

For simplicity, we assume $\hat{h} = \hat{e}^{(1)}$, i.e., the first principal direction of the dipole tensor at the saddle point is along the jump direction. This is not a bad assumption in most cases because $\hat{h}$ is in general along a direction of high symmetry of the saddle-point configuration. At any rate, modification of the analysis to the general case is straightforward.

Evaluation of integrals of the type *

$$\gamma(i_1, i_2, \dots, i_n) = \int \frac{d\Omega}{4\pi} \hat{r}_{i_1} \hat{r}_{i_2} \dots \hat{r}_{i_n} \quad (23)$$

constitutes the most difficult part of the calculation. These integrals can be expressed as a sum of products of Kronecker delta functions. A general expression for $\lambda(i_1, \dots, i_n)$ would be made up of $n!/2^{n/2}(n/2)!$ terms. In the present case the highest n involved is $n = 10$, giving a total of 945 terms. Fortunately, the presence of repeated indices makes it possible to reduce the number of terms drastically. Induction formulae can then be used to calculate these integrals. A more detailed account of the evaluation of these integrals is given by Woo [23].

From eq. (18), we obtain [23]

$$\langle \beta E_s \rangle = - \frac{\beta P Q k_0}{r^3} \frac{4}{15} (1 + \nu)(1 - p_1) \quad (24)$$

and

$$\langle \beta^2 E_s^2 \rangle = \frac{\beta^2 P^2 Q^2 k_0^2}{r^6} \left[\frac{16}{15} (2\nu - 1)^2 + C_1(p_\alpha p_\alpha - 3) + C_2(p_1^2 - 1) + C_3(p_1 - 1) \right], \quad (25)$$

where

$$C_1 = \frac{11000\nu^2 - 7568\nu + 7760}{17325},$$

$$C_2 = \frac{2112\nu^2 + 4224\nu + 42720}{17325},$$

$$C_3 = \frac{25344\nu^2 - 82368\nu - 16272}{17325}.$$

Note that $\langle \beta E_s \rangle$ vanishes for the isotropic case when $p_1 = 1$. It can be seen from eq. (21) that this is the correct limit. We also note that the sign of $\langle \beta E_s \rangle$ depends on the sign of Q and hence on whether the loop is vacancy or interstitial in character (see eq. (18)). If for one type of loop $\langle \beta E_s \rangle$ constitutes an attractive contribution to the interaction potential, then it would be a repulsive contribution for the other type. It is $\langle \beta E_s \rangle$

that distinguishes between the magnitudes of the point-defect current into the two types of loops. Such a distinction vanishes with $\langle \beta E_s \rangle$, as the point-defect shape becomes isotropic at the saddle point. For $\langle \beta^2 E_s^2 \rangle$, when $p_\alpha = 1$, only the first term in the square brackets in eq. (25) remains. That this is correct can be checked against the isotropic case, using eq. (21). Substituting into eq. (15), we obtain an expression for $\beta\bar{E}_s$:

$$\begin{aligned} \beta\bar{E}_s = & -\frac{4}{3}(1 + \nu)(1 - p_1) \frac{\beta P Q k_0}{r^3} \\ & - \left\{ \frac{8}{3}(1 - 2\nu)^2 + \frac{1}{2}C_1(p_\alpha p_\alpha - 3) \right. \\ & + \frac{1}{2}C_2(p_1^2 - 1) + \frac{1}{2}C_3(p_1 - 1) \\ & \left. - \frac{8}{25}(1 + \nu)^2(1 - p_1)^2 \right\} \frac{\beta^2 P^2 Q^2 k_0^2}{r^6}. \end{aligned} \quad (26)$$

Making the transformation

$$t = |\beta P Q k_0|^{-1/3} r \quad (27)$$

eq. (26) becomes

$$\beta\bar{E}_s(t) = A t^{-3} - B t^{-6} \quad (28)$$

where

$$A = \frac{4}{3}(1 + \nu)(1 - p_1) \text{sign}(PQ), \quad (29a)$$

$$\begin{aligned} B = & 1.6(1 - 2\nu)^2 + 1.5C_1(p_\alpha p_\alpha - 3) + 1.5C_2(p_1^2 - 1) \\ & + 1.5C_3(p_1 - 1) - 0.32(1 + \nu)^2(1 - p_1)^2 \end{aligned} \quad (29b)$$

and

$$\text{sign}(f) = \begin{cases} +1 & \text{for } f > 0 \\ -1 & \text{for } f < 0 \end{cases}$$

Note that A vanishes and only the first term of B survives when the point defect has an isotropic configuration at the saddle point. It is gratifying that eq. (28) indeed reduces to the isotropic limit which can be easily calculated from eq. (21).

4. The reaction radius and bias

Substitution of eq. (28) into eq. (13) gives the reaction radius for the constant boundary concentration method:

$$R_a = |k_0 \beta P Q|^{1/3} \left[\int_{r_a}^{r_s} t^{-2} \exp(A t^{-3} - B t^{-6}) dt \right]^{-1} \quad (30)$$

where

$$t_a = |k_0 \beta P Q|^{-1/3} r_a \quad (31a)$$

* Note that repeated arguments in γ are not summed over.

and

$$t_R = |k_0 \beta PQ|^{-1/3} R. \quad (31b)$$

Similarly, for the effective medium approximation, we have from eq. (14)

$$R_a = |k_0 \beta PQ|^{1/3} \left[\int_{t_\sigma}^{t_R} (t^{-2} - tR^{-3}) \times \exp(At^{-3} - Bt^{-6}) dt \right]^{-1}. \quad (32)$$

In the limit of low sink density, i.e., $R \rightarrow \infty$, expressions (30) and (32) become equal. For simplicity, we only examine this case in the following discussion. We believe the conclusions also apply for finite R values.

Before proceeding, let us consider the choice of the capture radius, r_σ , which defines a spherical surface on which the thermal equilibrium boundary condition is to

be supplied, and which encloses a region where the point defect can be considered to be captured by the sink. In other words, it is a region from which a newly arrived point defect will find it difficult to leave by thermal motion alone. It can only be annihilated at the sink. Outside this region, drift-diffusion takes place. In this regard, we follow Schroeder and Dettmann [19] and choose the capture region to be one bounded by the surface on which $E_g(\infty) - E_g(r) \approx \beta^{-1}$. However, such a region is not spherically symmetric and it is difficult to specify a capture radius. Fortunately, since the drift field ensures that most of the point-defect current enters the sink on the negative side of the drift potential, we may define the capture radius by requiring it to satisfy

$$E_g(\infty) - \min E_g(r_\sigma) = \beta^{-1}. \quad (33)$$

This definition may overestimate r_σ . Nevertheless, we shall see that the reaction radius calculated according to

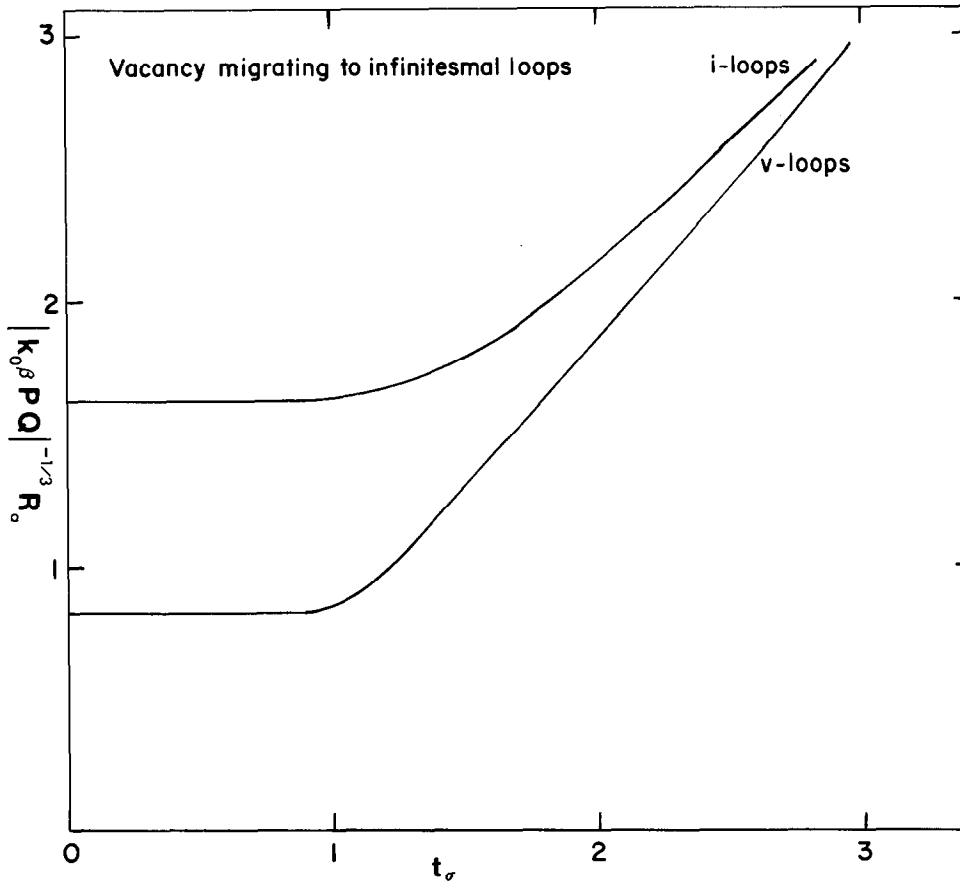


Fig. 1. Variation of $|k_0 \beta PQ|^{-1/3} R_a$ with t_σ (see eq. (31a)). Note the insensitivity of R_a when $t_\sigma < 1$.

eqs. (30) or (32) is not very sensitive to this overestimation, because the integrand is small for small values of t .

The drift field at the equilibrium configuration can be derived in the same way as eq. (18) and would have the same form, with P replaced by P_g , and p_σ by p_σ^g . The minimum occurs at $\hat{r} = \hat{b}$, or $\hat{r} \cdot \hat{b} = 0$, depending on the sign of $k_0 PQ$.

The two results differ by $\sim 25\%$, to which the value of R_a is insensitive (see fig. 1). It is a reasonable approximation to assume

$$t_\sigma \approx |P_g/P|^{1/3} \approx 1 \quad (34)$$

for most practical purposes. Thus, in the limit of low sink density, we may write

$$R_a = |k_0 \beta PQ|^{1/3} S^{-1}(A, B) \quad (35)$$

where $S(A, B)$ is defined by

$$S(A, B) = \int_1^\infty t^{-2} \exp(At^{-3} - Bt^{-6}) dt. \quad (36)$$

We can now calculate the bias B_{jL} of an IEDL:

$$B_{jL} = 1 - \frac{R_a^v}{R_a^i} \quad (37)$$

where $j = i$ for interstitial loops and v for vacancy loops, and the superscript on R_a denotes the type of point defect under consideration, i.e., i for interstitials and v for vacancies. From eq.(35) we obtain for vacancy loops

$$B_{vL} = 1 - \left| \frac{P_v}{P_i} \right|^{1/3} \frac{S(A_i^{vL}, B_i)}{S(A_v^{vL}, B_v)}, \quad (38)$$

and for interstitial loops

$$B_{iL} = 1 - \left| \frac{P_v}{P_i} \right|^{1/3} \frac{S(A_i^{iL}, B_i)}{S(A_v^{iL}, B_v)}. \quad (39)$$

The difference between the bias of a vacancy and an interstitial loop lies only in the difference between the A values. As can be seen from eqs. (29a) and (29b), we always have

$$A_j^{iL} = -A_j^{vL} \text{ and } j = i, v \quad (40)$$

because

$$Q^{iL} = -Q^{vL}.$$

Thus, for vacancy and interstitial loops to have the same bias, we must have $A_j^{iL} = 0 = A_j^{vL}$ for both vacancies and interstitials. From eq. (29a), this can only happen if both point-defect types have isotropic configurations at the saddle point.

When the saddle-point configuration is anisotropic, we would expect a difference between B_{iL} and B_{vL} . In

fact, we can write the difference $B_{iL} - B_{vL} \equiv \Delta B$ as

$$\Delta B = \left| \frac{P_v}{P_i} \right|^{1/3} \left[\frac{S(A_i^{vL}, B_i)}{S(A_v^{vL}, B_v)} - \frac{S(-A_i^{vL}, B_i)}{S(-A_v^{vL}, B_i)} \right] \quad (41)$$

where A_j^{vL} is given by eq. (29a) as

$$A_j^{vL} = \frac{4}{5}(1 + \nu)(1 - p_i^j) \text{ sign}(P_j) \quad (42)$$

where the superscript j denotes the type of defect: vacancy ($j = v$) or interstitial ($j = i$). We note that $S(A, B) - S(-A, B)$ has the same sign as A . Thus, from eq. (41), it can be seen that ΔB has the same sign as A_i^{vL} if $A_v^{vL} = 0$ or if A_i^{vL} and A_v^{vL} have opposite signs. On the other hand, ΔB has the opposite sign to A_v^{vL} if $A_i^{vL} = 0$. In the general case, when both A_v^{vL} and A_i^{vL} are non-zero, and have the same sign, the sign of ΔB then depends on the relative magnitudes of A_i^{vL} and A_v^{vL} . From this discussion, we see that vacancy loops and interstitial loops do have different biases even if their loop radii are the same [5,6]. The difference originates from the anisotropy of the saddle-point configuration; its sign and magnitude depend on the shape and the degree of distortion of the saddle-point configuration. All this information is contained in A_j^{vL} , which can be rewritten as

$$A^{vL} = \frac{4}{5}(1 + \nu) \frac{P_2 + P_3 - 2P_1}{|TrP|}. \quad (43)$$

It is negative for a defect elongated in the jump direction (cigar-shaped), and positive for one which is flattened in the jump direction (plate-shaped). The opposite is true for A^{iL} . The foregoing discussion can then be easily represented by the following picture. We divide point defects into three types: the cigar-shaped ones being M, the plate-shaped ones being F and the isotropic ones being N. The point-defect shape effect (SAPSE) can then be described as follows: relative to N-type defects, vacancy loops are more attractive to M-type defects and less attractive to F-type defects, and vice versa for interstitial loops. Thus, for example, if the vacancy is an F-type defect and the interstitial an M-type defect, the vacancy loop will have a higher bias (towards interstitials) than the interstitial loop. Or, if both the vacancy and the interstitial are of the same type, say F, and if the anisotropy of the interstitial is larger than that of the vacancy, then the vacancy loop will have a lower bias than the interstitial loop. Thus, the difference in the biases of the two types of loops is completely governed by the anisotropy of the point-defect shape at the saddle point. In fact, as can be seen from eqs. (38) and (39), the bias B is a function of the size factor, $|P_i/P_v|^{1/3}$, as well as the shape factor, $S(A_i,$

$B_i)/S(A_v, B_v)$. It is commonly believed that, since interstitials have a larger relaxation volume than vacancies, dislocation loops would always have a bias towards interstitials; whether it is a vacancy loop or an interstitial loop is immaterial (i.e., $B_{jL} > 0$ for $j = i, v$). Such reasoning is not necessarily valid when the shape factor is taken into account. The condition for $B > 0$, as derived from eqs. (38) and (39), is that

$$\frac{S(A_i, B_i)}{S(A_v, B_v)} < \left| \frac{P_i}{P_v} \right|^{1/3}. \quad (44)$$

If $|P_i/P_v|$ is large (say > 10), then eq. (44) is likely to be satisfied for both vacancy and interstitial loops. However, if $|P_i/P_v|$ is not so large (say < 3), then, depending on the shape anisotropy, eq. (44) may not be satisfied and it is possible to find that either the vacancy loop or the interstitial loop has a bias towards vacancies.

The physical reason for the intrinsically different biases for a vacancy loop and an interstitial loop, when the migrating point defect has an anisotropic shape at

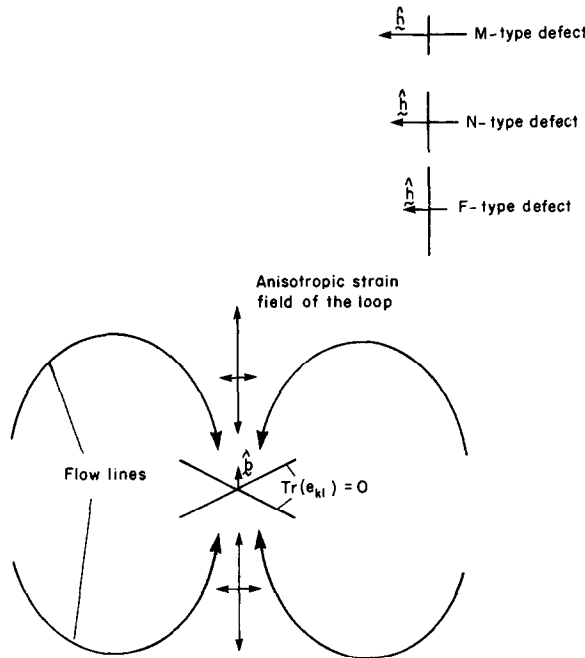


Fig. 2. Point-defect (with $P > 0$) migration to a vacancy dislocation loop showing the flow lines and the anisotropy of the loop strain field. Also shown are the dipole expansion fields of the three types of defects M, N, and F. Note the good "fit" of the M-type defects into the anisotropic strain field of the loop during the jump in the direction of the flow lines compared to the bad "fit" of the F-type defects.

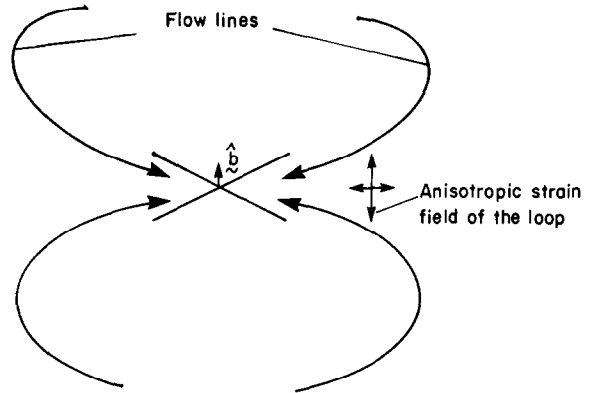


Fig. 3. Same as fig. 2, but for the case of an interstitial loop.

the saddle point, is clear from figs. 2 and 3. In fig. 2, we show the flow lines of point defects with a positive relaxation volume ($P > 0$) to a vacancy loop. Also shown is the anisotropic strain field (e_{kl}) of the loop in the region where the point defect enters the loop. It can be seen that e_{kl} is dilated more in the $\hat{b}$ direction than in the other directions. Consequently, an M-type defect would encounter a lower saddle-point energy barrier during its jump along the flow line (the most favoured jump), compared to an isotropic defect of the same relaxation volume. The opposite is true, of course, for an F-type defect. The case of an interstitial loop is shown in fig. 3. We note that, although the direction of the flow line is changed from along the $\hat{b}$ direction to perpendicular to the $\hat{b}$ direction, the loop strain field is still more dilated along the $\hat{b}$ direction. Thus, F-type defects, with larger dilatation perpendicular to the jump direction, would migrate faster than M-type defects with the same relaxation volume, which is opposite to the case for vacancy loops. The case for defects with negative relaxation volume ($P < 0$) can be similarly explained.

The foregoing discussion has a very significant implication for the rate-theory description of irradiation damage effects in crystalline materials. Accordingly, materials can be divided into three classes.

The first class contains materials in which both vacancy and interstitial loops have a bias towards interstitials, i.e., $B_{iL} > 0$ and $B_{vL} > 0$. These materials have a large P_i/P_v ratio. When irradiated by either MeV electrons or fast neutrons, void growth will occur. Free vacancy loops (i.e., free from the stress fields of other crystal defects) should not be observed. Examples of this class of materials may be copper, nickel, platinum, silver, gold, iron vanadium, niobium, molybdenum,

tantalum, magnesium and thallium [26] where void swelling occurs.

The second class contains materials in which interstitial loops have a bias towards interstitials, and vacancy loops have a bias towards vacancies, i.e., $B_{iL} > 0$ and $B_{vL} < 0$. These materials have a small P_i/P_v ratio and interstitials are more plate-like at the saddle point than vacancies. Void growth can only occur under MeV-electron irradiation, where vacancy loops are not nucleated (from cascade collapse). Under fast-neutron irradiation, where vacancy loops are created, the vacancy loops will grow and voids will not, because the former have a bias towards vacancies, whereas the latter have none. Examples of this class may be zirconium [26,27] and titanium [26].

The third class contains materials in which vacancy loops have a bias towards interstitials and interstitial loops have a bias towards vacancies. These materials also have a small P_i/P_v ratio, but vacancies are more plate-like than interstitials. Void growth does not occur under MeV-electron or fast-neutron irradiation if straight edge dislocations due to pre-irradiation deformation are absent. Dislocation loops will not occur because they cannot grow. Examples of this class of materials have not been observed. However, with the small number of elements or alloys on which irradiation experiments have been performed, not much can be said about this class, except that, if examples do exist, they may be useful nuclear structural materials.

Although the foregoing picture of the classification of materials may seem oversimplified in that there is no account of the influence of impurities, interactions of neighbouring defects, surface effects, stacking faults, dose rates and many other possible factors, it does provide a rough picture consistent with present knowledge to void growth and dislocation-loop growth in irradiated materials. It must be emphasized that the above discussion applies only to irradiations where a supersaturation of vacancies and interstitials is continuously maintained. In this case, the difference in point-defect arrival rates at the dislocation loops, due to a difference in the drift currents, is maintained during irradiation and causes profound microstructural effects. Such subtle drift-field effects are generally not important in processes which occur at point-defect concentrations close to their thermal equilibrium values.

5. The cases of fcc copper and bcc iron

To determine the magnitude of the effect discussed in the foregoing section, we apply the above theory to fcc copper and bcc iron. In these cases, the saddle-point dipole tensors of vacancies and interstitials have been estimated by computer simulation [20,21]. It is hoped that such an exercise can provide us with some information about the significance of the bias differential for the two kinds of loops.

Table 1

Parameters for vacancies and interstitials in the saddle-point configuration for fcc copper, obtained by computer simulation [20], to be used in the present study

Defect	Potentials	TrP (eV)	p_1	p_2	p_3	$\hat{e}^{(1)} (= \hat{h})$	$\hat{e}^{(2)}$	$\hat{e}^{(3)}$	Defect type (M or F)
Vacancy migration	M0	5.61	-0.73	0.21	3.52	1	1	0	F
						1	-1	0	
						0	0	1	
<100> Interstitial migration	M0	68.9	1.10	0.87	1.03	1	1	0	M
						0	0	1	
						1	-1	0	
Vacancy migration	M1	3.4	-1.13	0	4.13	1	1	0	F
						1	-1	0	
						0	0	1	
<100> Interstitial migration	M1	45.6	1.14	0.83	1.03	1	1	0	M
						0	-1	0	
						1	0	1	

Table 2

Parameters for vacancies and interstitials in the saddle-point configuration for bcc iron, obtained by computer simulation [21], to be used in the present study

Defect	Potentials	TrP (eV)	p_1	p_2	p_3	$\hat{e}^{(1)} (= \hat{h})$	$\hat{e}^{(2)}$	$\hat{e}^{(3)}$	Defect type (M or F)
Vacancy migration	Johnson	-2.82	2.73	0.135	0.135	1	1	1	F
						1	1	-1	
						1	-2	0	
$\langle 111 \rangle$ Interstitial migration	Johnson	63.1	1.50	0.89	0.61	1	1	1	M
						1.05	0	-1.90	
						1	-1	1	

In table 1, we list the parameters for vacancies and interstitials in the saddle-point configuration for fcc copper obtained by computer simulation [20]. Table 2 contains the corresponding information for iron [20,21].

The results for the biases B_{vL} and B_{iL} for the vacancy loop and the interstitial loop are given in table 3. For copper, dipole tensors corresponding to two interatomic potentials of the modified Morse type [20] give reasonably consistent results. For both metals, vacancy loops have a *significantly* higher bias for interstitials than interstitial loops. This is expected from the discussion in the previous section because the vacancies in this case are F-type defects and the interstitials are M-type defects (see tables 1 and 2). This is consistent with the fact that void growth, rather than vacancy loop growth, occurs in copper [26] and iron [26]. In fact, a vacancy loop in this case will absorb more interstitials than vacancies and start to shrink the moment it is created from cascade collapse. In effect, vacancy loops only act as recombination centres. The void-swelling behaviour of this material should be well described by theories that include the effect of vacancy-loop creation and shrinkage (see e.g., Bullough, Eyre and Kristian [9]).

Table 3

The biases of vacancy and interstitial loops in copper and iron (see text for meaning of symbols)

	Interatomic potential	B_{vL}	B_{iL}
Copper	M0	0.527	0.327
	M1	0.498	0.294
Iron	Johnson	0.604	0.445

6. Summary and conclusions

Point-defect migration into an infinitesimal edge dislocation loop in an isotropic linear elastic medium has been studied. Particular care has been taken to include the effects of the saddle-point shape anisotropy of the point defect. Expressions for the reaction radii and the biases have been derived. These are found to depend on parameters such as the loop strength, the loop nature (vacancy or interstitial), the relaxation volumes, and the shapes at the saddle point. From these expressions the following conclusions can be drawn:

1. For point defects with an anisotropic shape at the saddle point, there is an intrinsic difference between the reaction radii of a vacancy loop and an interstitial loop. In this context, point defects can be divided into M-type (elongated in the jump direction), F-type (elongated in the plane normal to the jump direction), and N-type (isotropic), according to their saddle-point configurations. The vacancy loops have the largest reaction radii for M-type defects and the smallest for F-type defects. The opposite is true for interstitial loops. This causes an intrinsic bias differential between vacancy loops and interstitial loops.

2. Application of the theory to copper and iron yields results that show that this bias differential can be quite significant.

3. Depending on the relative strengths of the size effect and the shape effect, it is not inconceivable that vacancy loops or interstitial loops with a negative bias (i.e., a preference for vacancies) exist in some materials.

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